Mahmoud Alyosify

Data Scientist

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SUMMARY

Data scientist who builds models around **the decision they need to support**. Predicted satellite collision risk 48+ hours ahead on highly imbalanced data, measured review effort on AI-written code, and designed evaluations with proper statistics (paired tests, bootstrap intervals). MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30); IBM AI Developer and AWS Cloud Practitioner certified; six years teaching probability and statistics.

SKILLS

Analysis: statistics, hypothesis testing, experiment design, A/B testing, feature engineering, imbalanced data
Modelling: scikit-learn, XGBoost, LightGBM, PyTorch, Optuna, SHAP
Data: SQL, Pandas, NumPy, PySpark, ETL pipelines, MySQL, MongoDB
Reporting: Power BI, Matplotlib, Streamlit, Crystal Reports
Programming: Python, R, SQL, Git

PROJECTS

SCRAP — satellite collision risk prediction 2026

Python, XGBoost, LightGBM, Optuna, SHAP

- Predicted satellite collision risk using only data available **48+ hours** before closest approach, so the warning arrives in time to act.
- Engineered features from raw conjunction records and handled a highly imbalanced target with metrics suited to rare, costly events.

Human vs. AI-Agent Pull Requests — research study

Python, GitHub data mining, statistics

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

MinimalLM — Master's thesis, published on arXiv 2026

Python, PyTorch, Hugging Face Transformers, NVIDIA A40

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).
- Paired statistics (McNemar tests, bootstrap intervals) and an LLM judge checked against a blind reading of 96 answers.

SimCLR-Vision-SSL — self-supervised image model 2026

PyTorch, SimCLR, SupCon, ONNX, FAISS

- Ran a controlled 42-experiment study to find which image augmentations actually improve the model; best result 84.30% top-1 accuracy.

RL-Cloud-Autoscaler — deep RL for cloud scaling 2026

Python, Gymnasium, PPO, DQN

- Built a cloud simulation (Gymnasium) and trained **PPO** and **DQN** agents to balance cost and latency, compared with standard threshold-based autoscaling.

Also: Diabetes Onset Prediction · ATM Transaction Analysis (R) · AudioShield (deepfake audio detection) · Horus-OSINT (Llama-3-8B fine-tuned on AWS).

EXPERIENCE

AI Engineer Intern Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) Maadi, Cairo

- Designed the evaluation: retrieval benchmarks (top-1, MRR), category-level scores and an LLM judge checked against a fixed metric (agreement 0.82).
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.

Systems Engineer (Military Service) Feb 2024 – Mar 2025
National Company for Roads Building & Development Cairo

- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and Crystal Reports.
- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.

Part-Time Machine Learning Instructor Jul 2025 – Present
National Telecommunication Institute (NTI) Egypt

- Teach **90-hour hands-on machine learning bootcamps**: regression, SVM, KNN, Naive Bayes, PCA and neural networks, in Python labs.
- Write the final exams and assess the trainees' end-to-end capstone projects.

Online Instructor 2020 – Present
Udemy & YouTube (Arabic courses) Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

CERTIFICATIONS

- IBM AI Developer Professional Certificate (IBM, 2026)
- Machine Learning Specialization (DeepLearning.AI & Stanford, 2026)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- AWS Certified Cloud Practitioner (Amazon Web Services)
- AWS Academy: Generative AI Foundations, Data Engineering, ML for NLP, ML Foundations

AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- Secretary, Higher Scientific & Technological Committee, Student Union, Assiut University

LANGUAGES

Arabic (native) · English (professional working proficiency)